LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 trial 1 seed 0

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

🧪 Great! I'll guide you through this explainability benchmarking study for the D(2) dopamine receptor. Let's start by finding a suitable dataset.

🔍 Step 1: Finding a Dataset

(Calling tool `find_datasets`)

Assistant

Excellent! I found the dataset `chembl_activity_data_P14416_binary.csv` which corresponds to the D(2) dopamine receptor (UniProt: P14416). Let's proceed with the workflow.

📊 Step 2: Loading the Dataset

(Calling tool `load_dataset`)

Assistant

(Calling tool `load_dataset`)

Assistant

🧬 Step 3: Computing Molecular Fingerprints

(Calling tool `compute_features`)

Assistant

✂️ Step 4: Splitting the Dataset

(Calling tool `split_dataset`)

Assistant

Now let's train the Deep Neural Network (DNN) model! ⚡

⚙️ Step 5: Training the DNN Model

(Calling tool `train_model`)

Assistant

The DNN model is training in the background. Let me check on its progress. ⏳

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

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Assistant

🎉 **Model Training Complete!** The DNN model achieved excellent performance:

Metric	Test Score
Accuracy	98.64%
AUC	0.9957
Balanced Accuracy	0.9864
F1 Score	0.9864
Precision	0.9849
Recall	0.9880

Now let's export predictions for the test set to identify a correctly predicted active compound.

📊 **Step 6: Exporting Predictions**

(Calling tool `export_predictions`)

Assistant

🎯 **Step 7: Selecting a Correctly Predicted Active Compound**

(Calling tool `select_compound_for_xai`)

Assistant

Perfect! I selected an active compound with 100% confidence:

- **SMILES**: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
- **True Label**: Active (1)
- **Predicted Label**: Active (1)
- **Confidence**: 1.0

Now let's run the **first pair of explainability methods: MolAnchor vs SHAP**.

🔬 **Pair 1: MolAnchor Analysis**

(Calling tool `explain_with_molanchor`)

Assistant

📊 **Pair 1: SHAP Analysis**

(Calling tool `explain_smiles_with_shap`)

Assistant

Let's visualize the SHAP values:

(Calling tool `plot_shap_mol`)

Assistant

Let's get the top contributing fingerprint bits from SHAP:

(Calling tool `get_top_k_bit_environments_from_shap`)

Assistant

🎭 **Pair 2: MolCE Analysis (R-group & Scaffold Analysis)**

(Calling tool `explain_with_molce`)

Assistant

🔄 **Pair 2: Counterfactual Analysis**

(Calling tool `generate_counterfactuals`)

Assistant

Let's visualize the counterfactuals:

(Calling tool `visualize_counterfactuals`)

Assistant

Now let me compile and summarize the comprehensive comparison between the two pairs of explainability methods.

📊 **COMPREHENSIVE EXPLAINABILITY COMPARISON SUMMARY**

🧪 Study Overview

- **Dataset**: ChEMBL D(2) dopamine receptor activity data (P14416)
- **Model**: Deep Neural Network (DNN)
- **Test Performance**: Accuracy 98.64%, AUC 0.9957
- **Selected Compound**: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
- **True Label**: Active (1) | **Predicted Label**: Active (1) | **Confidence**: 100%

🔬 Pair 1: MolAnchor vs SHAP Comparison

MolAnchor Results

Metric	Value
Anchor SMARTS	`[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`
Precision	1.0 (100%)
Coverage	0.53 (53%)
Interpretation	This is a **piperazine-based scaffold** with a carbonyl (C=O) and methylated nitrogen. The high precision indicates this fragment is *critical* for the model's active prediction.

🎯 **Key Structural Determinant Identified**: The **piperazine-carbonyl motif** is the primary anchor driving the active prediction.

SHAP Results

Rank	Fingerprint Bit	Contribution	Local Environment
1	1280	**+6.20**	`CCCN(C)C` (alkyl-amine chain)
2	1791	**+5.93**	`CN(C)C` (methylated amine)
3	1792	**+5.74**	`cc(CN(C)C)c` (aromatic-amine linkage)
4	885	**+5.39**	`cc(c)s` (thiophene/aromatic sulfur)
5	552	**-5.37**	`ccs` (negative contribution)

🎯 **Key Structural Determinants Identified**:

- **Positive drivers**: Alkyl-amine chains, methylated nitrogens, and aromatic-amine linkages
- **Negative driver**: Thiophene/aromatic sulfur environment

✅ AGREEMENTS Between MolAnchor & SHAP

1. **Nitrogen-containing motifs**: Both methods highlight **methylated/piperazine nitrogens** as critical. MolAnchor identifies the piperazine-carbonyl scaffold, while SHAP's top 3 bits all involve amine/alkyl-amine environments.
2. **Aromatic interactions**: SHAP's bit 1792 (`cc(CN(C)C)c`) and bit 885 (`cc(c)s`) point to aromatic-amine linkages, which align with MolAnchor's scaffold anchor that includes aromatic rings.
3. **Overall theme**: Both confirm that **amine-containing heterocycles** are the primary drivers of activity.

❌ DISAGREEMENTS Between MolAnchor & SHAP

1. **Granularity**: MolAnchor identifies a **single high-level scaffold** (piperazine-carbonyl), while SHAP reveals **multiple local environments** contributing independently (alkyl chains, methyl groups, aromatic linkages).
2. **Carbonyl emphasis**: MolAnchor heavily weights the **carbonyl (C=O)** as part of the anchor, but SHAP's top contributions don't explicitly highlight carbonyl environments—the model appears to care more about the **amine/nitrogen** chemistry than the carbonyl itself.
3. **Sulfur role**: SHAP shows a **negative contribution** from thiophene sulfur (bit 552: -5.37), suggesting sulfur-containing rings may *decrease* activity. MolAnchor doesn't capture this nuance, treating the scaffold more uniformly.

🎭 Pair 2: MolCE vs Counterfactuals Comparison

MolCE Results (R-group & Scaffold Analysis)

Type	Rank	Feature	Contrast Score	Interpretation
R-group	1	`*CC(C)NC(C)=O`	1.0	Substituent that *reinforces* activity
	2	`*C=C(F)C(=O)OCC`	1.0	Fluorinated ester substituent
	3	`*C(O)CNC(C)CC`	1.0	Hydroxy-amine chain
Scaffold	1	`O=C1N([*:1])CN(c2ccccc2)C12CCN(Cc1coc3ccccc13)CC2`	0.0	Current scaffold (no change needed)

🎯 **Key Insight**: The **R-group at position 1** is highly contrastive—replacing it with different substituents (even highly similar ones) strongly shifts predictions toward inactivity. The scaffold itself is stable and doesn't need changing.

Counterfactuals Results

Rank	Counterfactual SMILES	Change Type	Predicted Class	Probability (Inactive)	Tanimoto Similarity
1	`COCCCON1CN(...)`	Substituent	**Inactive** (0)	99.98%	0.683
2	`COC(C)C(N)N1CN(...)`	Substituent	**Inactive** (0)	87.76%	0.683
3	`COC(=O)CON1CN(...)`	Substituent	**Inactive** (0)	100.0%	0.672
4	`CN(C)CCCON1CN(...)`	Substituent	**Inactive** (0)	87.40%	0.662
5	`COC(=O)COCN1CN(...)`	Substituent	**Inactive** (0)	99.99%	0.662

🎯 **Key Insight**: **All counterfactuals are substituent changes** (no scaffold changes were needed). The **R-group at position 1** is the most sensitive region—adding polar groups (esters, ethers, fluorinated chains) or extending alkyl chains flips the prediction to inactive.

✅ AGREEMENTS Between MolCE & Counterfactuals

1. **R-group sensitivity**: Both methods identify the **R-group at position 1** as the critical region. MolCE shows high contrast scores for R-group substitutions, while counterfactuals all modify this same position.
2. **Polar/electronegative groups**: MolCE's top R-group `*CC(C)NC(C)=O` (amide-like) and counterfactuals with **esters** (`COC(=O)`) and **fluorinated chains** (`C=C(F)`) both point to **polarity/electronegativity** as key drivers.
3. **Scaffold stability**: MolCE reports scaffold contrast score = 0.0 (scaffold doesn't need changing), and counterfactuals confirm this—all successful counterfactuals are **substituent-only** changes.

❌ DISAGREEMENTS Between MolCE & Counterfactuals

1. **Direction of effect**: MolCE's R-group analysis shows *which* substituents *reinforce* activity (contrastive = toward foil class 0). Counterfactuals show *which* substituents *destroy* activity. MolCE focuses on **what makes it active**; counterfactuals focus on **what breaks it**.
2. **Specificity**: MolCE identifies **3 distinct R-group motifs** that are contrastive, while counterfactuals show **10+ different substituent patterns** that all work. MolCE is more selective; counterfactuals are more exhaustive.
3. **Sulfur role**: Counterfactuals don't explicitly highlight sulfur, while MolCE's analysis doesn't address it either—but SHAP (from Pair 1) shows sulfur is *negative*. This suggests the **thiophene ring** in the original molecule is tolerated but not beneficial.

🎯 OVERALL CROSS-METHOD SYNTHESIS

Where ALL Methods Agree ✅

Structural Feature	Evidence From
Piperazine/methylated amine	MolAnchor (anchor scaffold), SHAP (bits 1280, 1791), MolCE (R-group analysis)
Aromatic-amine linkage	MolAnchor (scaffold includes aromatic), SHAP (bit 1792), Counterfactuals (substituent changes at this position)
R-group position 1 sensitivity	MolCE (high contrast), Counterfactuals (all CFs modify this position)
Necessity of nitrogen heterocycles	All methods point to nitrogen-containing motifs as essential

Where Methods Diverge ❌

Aspect	MolAnchor/SHAP	MolCE/Counterfactuals
Focus	*What makes it active* (positive features)	*What breaks it* (perturbation sensitivity)
Granularity	Local atom/bit-level	R-group/scaffold-level
Sulfur role	SHAP: **Negative** contribution	MolCE/CF: **Neutral/tolerated**
Carbonyl role	MolAnchor: **Critical** anchor	SHAP: **Not top contributor**
Actionability	Identifies *key fragments*	Identifies *modifiable positions*

📌 Key Takeaways for Drug Design

1. **Keep the piperazine-carbonyl core** (MolAnchor confirms it's critical; SHAP shows amines drive activity).

2. **Optimize the R-group at position 1** (MolCE + Counterfactuals agree this is the most tunable region for potency).
3. **Avoid thiophene/sulfur** (SHAP shows negative contribution—consider replacing with benzene or pyridine).
4. **Polarity matters** (Counterfactuals show polar substituents at R1 flip activity—balance lipophilicity carefully).

This multi-method consensus provides a **robust, triangulated understanding** of the D(2) receptor pharmacophore that no single method could achieve alone! 🧬 ✨